# Vectors

## Vector Basics

**Definition:** A **vector** has both magnitude (size) and direction. A **scalar** has only magnitude. Vectors are written as $\mathbf{a}$, $\vec{AB}$, or in column form $\begin{pmatrix} x \\ y \end{pmatrix}$.

In 2D, a vector can be written as $\mathbf{a} = x\mathbf{i} + y\mathbf{j} = \begin{pmatrix} x \\ y \end{pmatrix}$.

In 3D, a vector can be written as $\mathbf{a} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$.

### Magnitude

$$|\mathbf{a}| = \sqrt{x^2 + y^2} \quad (2\text{D}) \qquad |\mathbf{a}| = \sqrt{x^2 + y^2 + z^2} \quad (3\text{D})$$

### Unit Vectors

**Unit Vector:** A vector of magnitude 1 in the direction of $\mathbf{a}$:

$$\hat{\mathbf{a}} = \frac{\mathbf{a}}{|\mathbf{a}|}$$

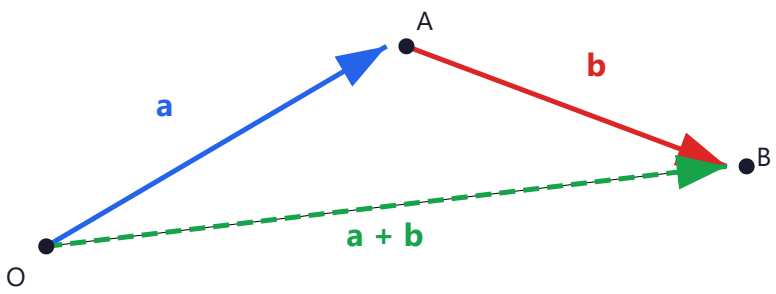
## Vector Operations

### Addition and Subtraction

Vectors are added component-wise:

$$\begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} + \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} = \begin{pmatrix} a_1 + b_1 \\ a_2 + b_2 \\ a_3 + b_3 \end{pmatrix}$$

Triangle Law of Addition




## Scalar Multiplication

Multiplying vector $\mathbf{a}$ by scalar $k$ scales its magnitude by $|k|$. If $k < 0$, the direction reverses.

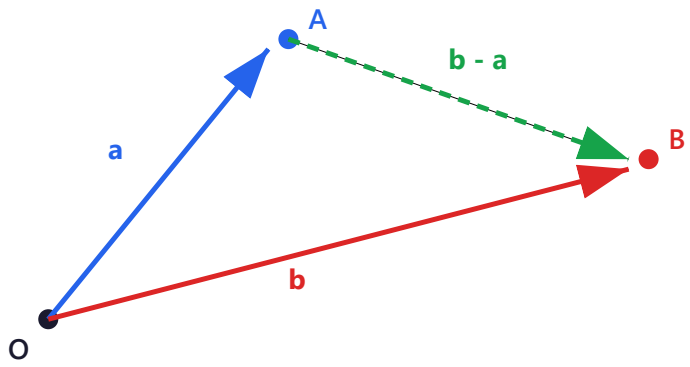
# Position Vectors

**Position Vector:** The position vector of a point $A$ is $\vec{OA}$, the vector from the origin $O$ to $A$. If $A = (a_1, a_2, a_3)$, then $\vec{OA} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$.

The vector from $A$ to $B$ is:

$$\vec{AB} = \vec{OB} - \vec{OA} = \mathbf{b} - \mathbf{a}$$

Position Vectors: AB = b - a




### Worked Example: Distance Between Points

Find the distance between $A(3, 1, -2)$ and $B(1, -1, 4)$.

**Solution:**

$$\vec{AB} = \begin{pmatrix} 1-3 \\ -1-1 \\ 4-(-2) \end{pmatrix} = \begin{pmatrix} -2 \\ -2 \\ 6 \end{pmatrix}$$

$$|\vec{AB}| = \sqrt{(-2)^2 + (-2)^2 + 6^2} = \sqrt{4+4+36} = \sqrt{44} = 2\sqrt{11}$$

## Scalar Product (Dot Product)

**Scalar Product:** For vectors $\mathbf{a}$ and $\mathbf{b}$:

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}| \cos\theta$$

where $\theta$ is the angle between them. In component form:

$$\mathbf{a} \cdot \mathbf{b} = a_1 b_1 + a_2 b_2 + a_3 b_3$$

Key properties:

- $\mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{a}$ (commutative)
- If $\mathbf{a} \cdot \mathbf{b} = 0$, the vectors are **perpendicular**
- $\mathbf{a} \cdot \mathbf{a} = |\mathbf{a}|^2$

## Angle Between Two Vectors

$$\cos\theta = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}||\mathbf{b}|}$$

### Worked Example: Angle Between Vectors

Find the angle between $\mathbf{a} = \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 1 \\ -1 \\ 3 \end{pmatrix}$.

**Solution:**

$$\mathbf{a} \cdot \mathbf{b} = (2)(1) + (1)(-1) + (-1)(3) = 2 - 1 - 3 = -2$$

$$|\mathbf{a}| = \sqrt{4 + 1 + 1} = \sqrt{6}, \quad |\mathbf{b}| = \sqrt{1 + 1 + 9} = \sqrt{11}$$

$$\cos\theta = \frac{-2}{\sqrt{6}\sqrt{11}} = \frac{-2}{\sqrt{66}} \implies \theta = \cos^{-1}\left(\frac{-2}{\sqrt{66}}\right) = 104.3^\circ$$

**Exam Tip:** If the dot product is negative, the angle is obtuse (greater than 90 degrees). Always give angles to 1 decimal place unless told otherwise. Remember that the angle between two vectors is always between 0 and 180 degrees.

## Geometric Problems with Vectors

### Midpoint

The midpoint $M$ of $AB$ has position vector:

$$\vec{OM} = \frac{1}{2}(\mathbf{a} + \mathbf{b})$$

### Dividing a Line Segment

The point dividing $AB$ in the ratio $m : n$ has position vector:

$$\vec{OP} = \frac{n\mathbf{a} + m\mathbf{b}}{m + n}$$

### Worked Example: Proving Perpendicularity (with coordinates)

Points $A(1, 2, 3)$, $B(3, 0, 1)$, and $C(4, 2, 0)$. Show that angle $ABC = 90^\circ$.

**Solution:** Find vectors from $B$:

$$\vec{BA} = \begin{pmatrix} 1-3 \\ 2-0 \\ 3-1 \end{pmatrix} = \begin{pmatrix} -2 \\ 2 \\ 2 \end{pmatrix}, \quad \vec{BC} = \begin{pmatrix} 4-3 \\ 2-0 \\ 0-1 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$$

$$\vec{BA} \cdot \vec{BC} = (-2)(1) + (2)(2) + (2)(-1) = -2 + 4 - 2 = 0$$

Since $\vec{BA} \cdot \vec{BC} = 0$, the vectors are perpendicular, so angle $ABC = 90^\circ$.

### Worked Example: Proving Perpendicularity

Show that $\mathbf{p} = 3\mathbf{i} + \mathbf{j} - 2\mathbf{k}$ and $\mathbf{q} = 2\mathbf{i} - 4\mathbf{j} + \mathbf{k}$ are perpendicular.

**Solution:**

$$\mathbf{p} \cdot \mathbf{q} = (3)(2) + (1)(-4) + (-2)(1) = 6 - 4 - 2 = 0$$

Since $\mathbf{p} \cdot \mathbf{q} = 0$, the vectors are perpendicular.

**Exam Tip:** To prove vectors are parallel, show that one is a scalar multiple of the other. To prove perpendicularity, show the dot product is zero. These are the two most common vector proof questions at A-Level.

## Vector Equation of a Line

**Vector Equation of a Line:** A straight line through point $A$ with position vector $\mathbf{a}$, in direction $\mathbf{d}$, has equation:

$$\mathbf{r} = \mathbf{a} + \lambda \mathbf{d}$$

where $\lambda$ is a scalar parameter. Every point on the line corresponds to a unique value of $\lambda$.

To write the line through two points $A$ and $B$:

- Direction vector: $\mathbf{d} = \vec{AB} = \mathbf{b} - \mathbf{a}$
- Line equation: $\mathbf{r} = \mathbf{a} + \lambda(\mathbf{b} - \mathbf{a})$

### Worked Example: Equation of a Line

Find the vector equation of the line through $A(2, 1, -3)$ and $B(4, 0, 1)$.

**Solution:**

$$\vec{AB} = \begin{pmatrix} 4-2 \\ 0-1 \\ 1-(-3) \end{pmatrix} = \begin{pmatrix} 2 \\ -1 \\ 4 \end{pmatrix}$$

$$\mathbf{r} = \begin{pmatrix} 2 \\ 1 \\ -3 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -1 \\ 4 \end{pmatrix}$$

## Checking if a Point Lies on a Line

A point $P$ lies on $\mathbf{r} = \mathbf{a} + \lambda \mathbf{d}$ if there exists a consistent value of $\lambda$ satisfying all three component equations simultaneously.

### Worked Example: Point on a Line

Does $C(8, -2, 13)$ lie on $\mathbf{r} = \begin{pmatrix} 2 \\ 1 \\ -3 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -1 \\ 4 \end{pmatrix}$?

**Solution:** From the $x$-component: $2 + 2\lambda = 8 \implies \lambda = 3$.

Check $y$: $1 + (-1)(3) = -2$. ✓

Check $z$: $-3 + 4(3) = 9 \neq 13$. ✗

The $\lambda$ values are inconsistent, so $C$ does not lie on the line.

## Collinearity

Three points $A$, $B$, $C$ are **collinear** if $\vec{AC} = k\,\vec{AB}$ for some scalar $k$ — i.e. the vectors are parallel and share a common point.

### Worked Example: Proving Collinearity

Show that $A(1, 2, 3)$, $B(3, 5, 7)$ and $C(5, 8, 11)$ are collinear.

**Solution:**

$$\vec{AB} = \begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix}, \quad \vec{AC} = \begin{pmatrix} 4 \\ 6 \\ 8 \end{pmatrix} = 2\begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix} = 2\,\vec{AB}$$

Since $\vec{AC} = 2\,\vec{AB}$, the vectors are parallel and share point $A$, so $A$, $B$, $C$ are collinear.

## Intersection and Skew Lines

In 3D, two lines can either **intersect**, be **parallel**, or be **skew** (non-parallel and non-intersecting).

To test two lines $\mathbf{r} = \mathbf{a} + \lambda\mathbf{d}$ and $\mathbf{r} = \mathbf{b} + \mu\mathbf{e}$:

1. Set $\mathbf{a} + \lambda\mathbf{d} = \mathbf{b} + \mu\mathbf{e}$ and write out three component equations
2. Solve two of the equations for $\lambda$ and $\mu$
3. Substitute into the third equation — if satisfied, the lines intersect; otherwise they are skew

**Exam Tip:** Always check whether direction vectors are scalar multiples of each other first — if they are, the lines are parallel (or the same line). Only if they are not parallel do you need to test for intersection vs. skew.

## Angle Between Two Lines

The acute angle $\theta$ between lines with direction vectors $\mathbf{d}_1$ and $\mathbf{d}_2$:

$$\cos\theta = \frac{|\mathbf{d}_1 \cdot \mathbf{d}_2|}{|\mathbf{d}_1||\mathbf{d}_2|}$$

The modulus signs ensure the acute angle is always returned.

### Worked Example: Angle Between Lines

Find the acute angle between lines with directions $\mathbf{d}_1 = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$ and $\mathbf{d}_2 = \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix}$.

**Solution:**

$$\mathbf{d}_1 \cdot \mathbf{d}_2 = 3 - 2 - 2 = -1$$

$$|\mathbf{d}_1| = \sqrt{6}, \quad |\mathbf{d}_2| = \sqrt{14}$$

$$\cos\theta = \frac{|-1|}{\sqrt{6}\sqrt{14}} = \frac{1}{\sqrt{84}} \implies \theta = \cos^{-1}\left(\frac{1}{\sqrt{84}}\right) \approx 83.7^\circ$$